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Systematic Review

Standardization and Benchmarking of Datasets and Evaluation Protocols for Machine Learning-Based Water Leak Detection: A Systematic Review

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Abstract

Despite the increasing use of machine learning (ML) and deep learning (DL) for leak detection and localization in water distribution networks (WDNs), standardized datasets, common evaluation protocols, and consistent reporting practices remain limited, thereby restricting meaningful comparison among studies. In this systematic review, PRISMA 2020 was followed to examine the benchmarking practices, data sources, sensing modalities, and evaluation strategies adopted in recent ML/DL-based leak-detection studies. Scopus was searched in April 2026, and 90 studies were included after screening against predefined eligibility criteria. Substantial variation was identified in dataset generation, model validation, and performance assessment. Thirty studies were assigned to the public-benchmark category, but only 23 used one of the three named water-leak resources tracked in this review; the remainder included topology-only reuse or other heterogeneous public resources. Pressure measurements were the dominant sensing modality, whereas flow, transient, and smart-meter data remained comparatively underexplored. Considerable inconsistency was also observed in performance metrics, validation procedures, and leak-scenario definitions, preventing fair comparison of detection and localization performance. Based on these findings, a practical reporting framework is proposed to improve transparency, reproducibility, and comparability. The principal recommendations include standardized dataset documentation, consistent evaluation protocols, expanded community benchmarks, uncertainty quantification, and model interpretability to support operational deployment.

Keywords: water distribution networks; leak detection; machine learning; deep learning; benchmarking; datasets; evaluation protocols; standardization; reproducibility

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1. Introduction

Water lost between treatment and the customer is one of the largest and most persistent inefficiencies in urban infrastructure. In the United States and Canada alone, about 260,000 water mains fail each year, at a rate of 11.1 breaks per 100 miles of pipe, and roughly a fifth of the installed pipe stock is judged to need replacement [1]. Losses of 20% to 30% of supplied water are common, and higher figures are reported in aging or poorly monitored networks; recent Italian studies put average network losses at around 40% of the water transported, with the associated economic and energy costs [2,3]. Every liter lost has already been abstracted, treated, and pumped, so leakage carries an energy and emissions

penalty as well as a financial one. Against this background, leak detection and localization have received considerable research attention, and over the past decade the emphasis has moved from hardware-based surveys and hydraulic model calibration towards data-driven methods based on machine learning (ML) and deep learning (DL) [4,5].

This shift has produced a large volume of algorithmic work. Convolutional neural networks (CNNs) applied to pressure residual maps [6,7], one-dimensional CNN autoencoders operating on multivariate SCADA time series [8], Long Short-Term Memory (LSTM)-based autoencoders for unsupervised anomaly detection [9–11], graph neural networks that exploit network topology [12–14], two-stage detection and localization pipelines [15], generative models addressing data scarcity [7,16], and, more recently, vision-language and foundation-model approaches [17] have all been proposed for the leak detection and localization task. Comparative surveys confirm both the breadth of this activity and its increasing pace [4,5,18].

A basic problem complements this progress: the field has no shared basis for comparison. Almost every new model is trained and evaluated on a different dataset. In many cases this is a bespoke synthetic scenario set generated with EPANET on a small benchmark topology, and in others a proprietary utility dataset that cannot be released. The evaluation metrics, leak scenario definitions, and validation schemes are chosen by the authors. As was observed by the organizers of the Battle of the Leakage Detection and Isolation Methods (BattLeDIM) competition, a key challenge in the design of leak detection algorithms is the evaluation and comparison of performance between methodologies using common benchmarks [19]. When two papers each report, for example, an accuracy above 95%, the numbers are rarely comparable. They may refer to different network topologies, leak sizes, sensor densities, noise assumptions, class balances, and even to different definitions of what counts as a correct localization.

Previous reviews have documented the methods available. A criteria-based critical review of AI applications in water-leak management was presented by Abdelmageed et al. [4]; AI applications were surveyed comparatively by Kammoun et al. [5]; and an earlier comprehensive review by Farah and Shahrour [2] mapped hardware-, software-, and smart-network approaches, while addressing dataset provenance only partially and not analyzing benchmark reuse. Transient-based methods were reviewed with a statistical and methodological focus by Ayati et al. [20], and ML/DL models, input variables, databases, and output strategies were systematically reviewed by Zuñiga-Uribe et al. [18]. Table 1 summarizes the scope of these five prior reviews. Unlike the earlier work, the present review treats dataset provenance, benchmark reuse, and evaluation protocols as the principal objects of analysis across the 90-study corpus and uses the resulting evidence to propose a minimum reporting framework.

The specific objectives of the review are as follows:

1. To map the datasets used in ML/DL water leak detection research, distinguishing public benchmarks, bespoke synthetic data, laboratory testbeds, and proprietary field data (RQ1);
2. To assess the evaluation protocols that are applied, including metrics, validation schemes, leak scenario definitions, and baselines, and to quantify their variability (RQ2);
3. To determine the state of reproducibility, including data and code availability and the reuse of common benchmarks (RQ3);
4. To derive a minimum reporting standard and a benchmarking roadmap for the field (RQ4).

Table 1. Scope of prior reviews of ML/DL-based water leak detection.

Review	Corpus	Primary Focus	Dataset Provenance	Benchmark Reuse	Evaluation Protocols	Reporting Standard
Abdelmageed et al. [4]	Not stated	Criteria-based critical review of AI applications in water-leak management	Not analyzed	Not analyzed	Partial	Not proposed
Kammoun et al. [5]	Selective	Comparative survey of AI algorithms for leak detection	Not analyzed	Not analyzed	Partial	Not proposed
Farah and Shahrour [2]	600 records	Leak detection methods: hardware, software, smart networks	Partial	No	Partial	No
Ayati et al. [20]	139 records	Transient-based methods, statistical and methodological focus	Discussed per study	Not analyzed	Discussed per study	Not proposed
Zuñiga-Uribe et al. [18]	53 studies	ML/DL models, input variables, databases, and output strategies	Discussed per study	Discussed per study	Discussed per study	Not proposed

2. Materials and Methods

2.1. Search Strategy, Screening and Eligibility

This review was reported in accordance with PRISMA 2020 [21]. The search concepts, eligibility criteria, extraction fields, and synthesis procedures were specified before screening; however, the protocol was not prospectively registered. Scopus was searched in April 2026 using four concept blocks: (i) leak detection and localization; (ii) water distribution networks and pipes; (iii) machine learning, deep learning, and artificial intelligence; and (iv) dataset and benchmarking terms (dataset, benchmark, and training data). Titles, abstracts, and keywords were searched. Scopus was selected because it is a major multidisciplinary abstract and citation database with broad coverage of peer-reviewed water-engineering journals and computer-science conference proceedings relevant to this interdisciplinary field. Its consistent citation indexing and bibliographic metadata also support a reproducible search and bibliometric analysis. Nevertheless, complete coverage of the evidence base cannot be guaranteed by a single-database search; this limitation is acknowledged in Section 8.1. No language restriction was applied in the search string. No date restriction was applied other than the April 2026 search cutoff, and no document-type restriction was applied beyond the eligibility criteria stated in items (a) to (c) below.

The search identified 102 records. After verification, no duplicate records and no records with missing abstracts were found. The documents were classified by type: 62 journal articles (60.8%), 34 conference papers (33.3%), 3 reviews (2.9%), 1 data paper, 1 book chapter, and 1 conference review. Conference papers were retained because benchmark datasets and competition results in this field are frequently introduced at conference venues [6,19]. Screening and eligibility assessment were performed manually by the authors. To be included, a study was required to (a) address leak detection, localization, or leakage-related anomaly detection in water distribution networks or water pipelines; (b) apply ML or DL methods, or contribute a dataset or benchmark for such methods; and (c) provide a sufficient methodological description of the data used. Exclusion criteria were coded as follows. E1 was assigned to studies outside the water domain, such as purely oil and gas or underwater oil applications. E2 was assigned to studies that did not address leakage, such as water quality monitoring and source contamination. E3 was assigned where no ML or DL methodology was applied. E4 was assigned to documents that were not primary studies, including conference front matter, editorial material, and book chapters without an original study. E5 was assigned to review articles, which were retained as background literature but excluded from the quantitative synthesis. Twelve

records were excluded: 2 under E1 (underwater oil leakage detection; fault detection in subsea pipelines), 5 under E2 (water quality monitoring; source contamination detection; weld-seam leak identification in pipe manufacturing; a dynamic leakage risk assessment framework; and non-revenue water predictive modeling), 2 under E4 (conference proceedings front matter; a book chapter on big data and carbon neutrality), and 3 under E5 (the review articles [4,18,20]). The screening therefore yielded 90 included studies (Figure 1).

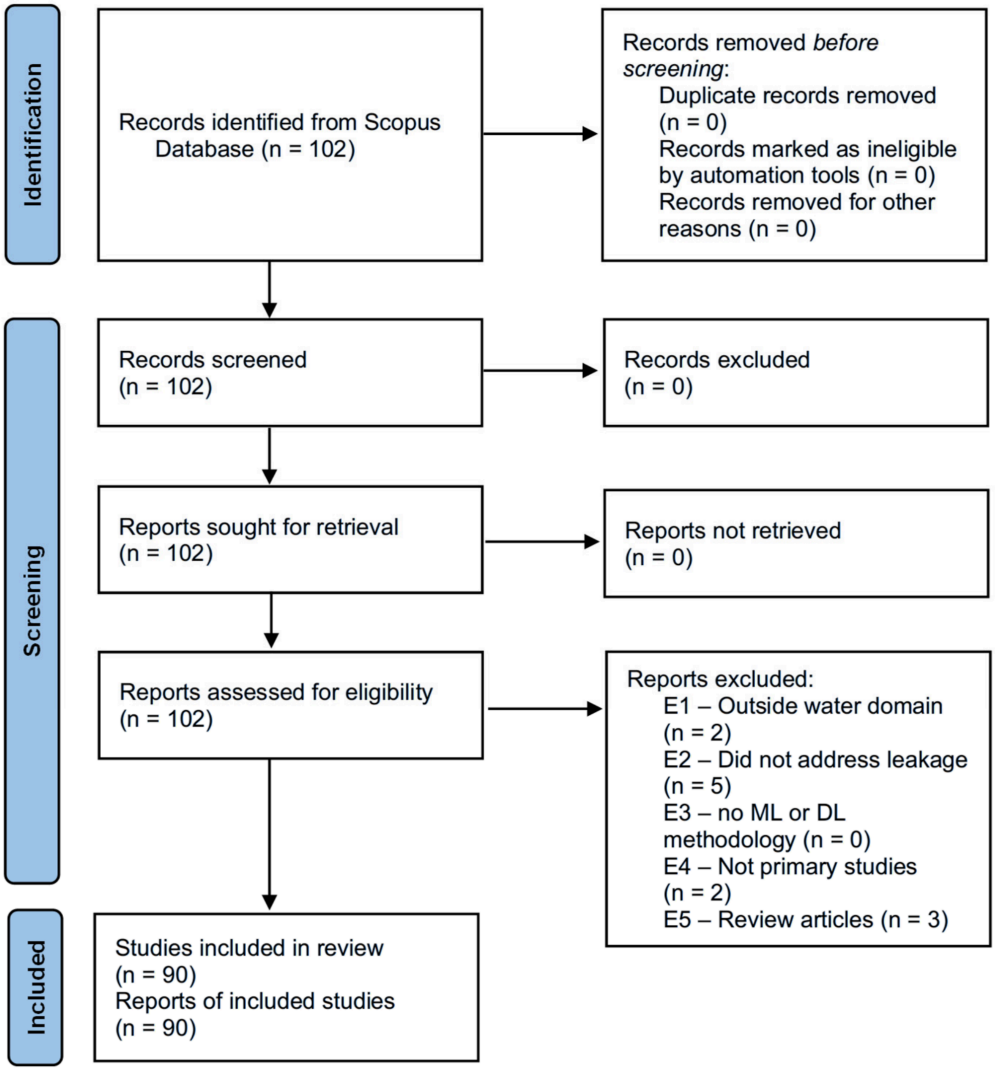


Figure 1. PRISMA 2020 flow diagram of the identification, screening, and inclusion process.

2.2. Data Extraction and Synthesis

The literature corpus for this review builds upon our prior comprehensive review [2], with additional records identified through the updated Scopus search described above. For each included study, the following fields were extracted from titles, abstracts, and author keywords, and were complemented by full-text verification where this was available: data source category (public benchmark, bespoke synthetic or EPANET-based, laboratory or testbed, field or utility data); sensing modality (pressure, flow, acoustic or vibration, transient, smart meter); ML/DL model family; use of named benchmarks (LeakDB, BattLeDIM, L-Town, Hanoi, Net1, and similar); evaluation metrics and validation scheme; and indicators of reproducibility (data availability, code availability, open access status). Thematic coverage was quantified through structured keyword mapping, and descriptive statistics were computed for the bibliometric characterization. All quantitative analyses, comprising

the bibliometric characterization and the dataset and evaluation-protocol synthesis, were performed on the studies retained after PRISMA screening (n = 90), so that every reported figure refers to the included evidence base. In addition, an author-keyword co-occurrence network was constructed after term normalization, in which singular and plural forms and synonym variants were merged. Keywords with at least two occurrences were retained, and thematic clusters were detected through greedy modularity optimization, following the VOSviewer v 1.6.20 co-occurrence mapping workflow. All 90 studies retained after screening are cited in the reference list.

3. Bibliometric Overview of the Included Studies

3.1. Temporal Evolution

A pronounced growth trajectory is exhibited by the included studies (Figure 2). Only 8 of the 90 studies (8.9%) were published before 2022 (2019: 2; 2020: 3; 2021: 3), after which output accelerates sharply, with 7 studies in 2022, 13 in 2023, 19 in 2024, 24 in 2025, and 19 in the first four months of 2026 alone. Nearly 70% of the included studies (68.9%) were published in 2024 or later. This distribution confirms that dataset-aware and benchmark-aware ML/DL leak detection is a young and rapidly expanding research area, and it coincides with the release and uptake of the first community benchmarks of the field, namely LeakDB and the BattLeDIM competition dataset, whose summary paper [19] is the most cited study in the included database.

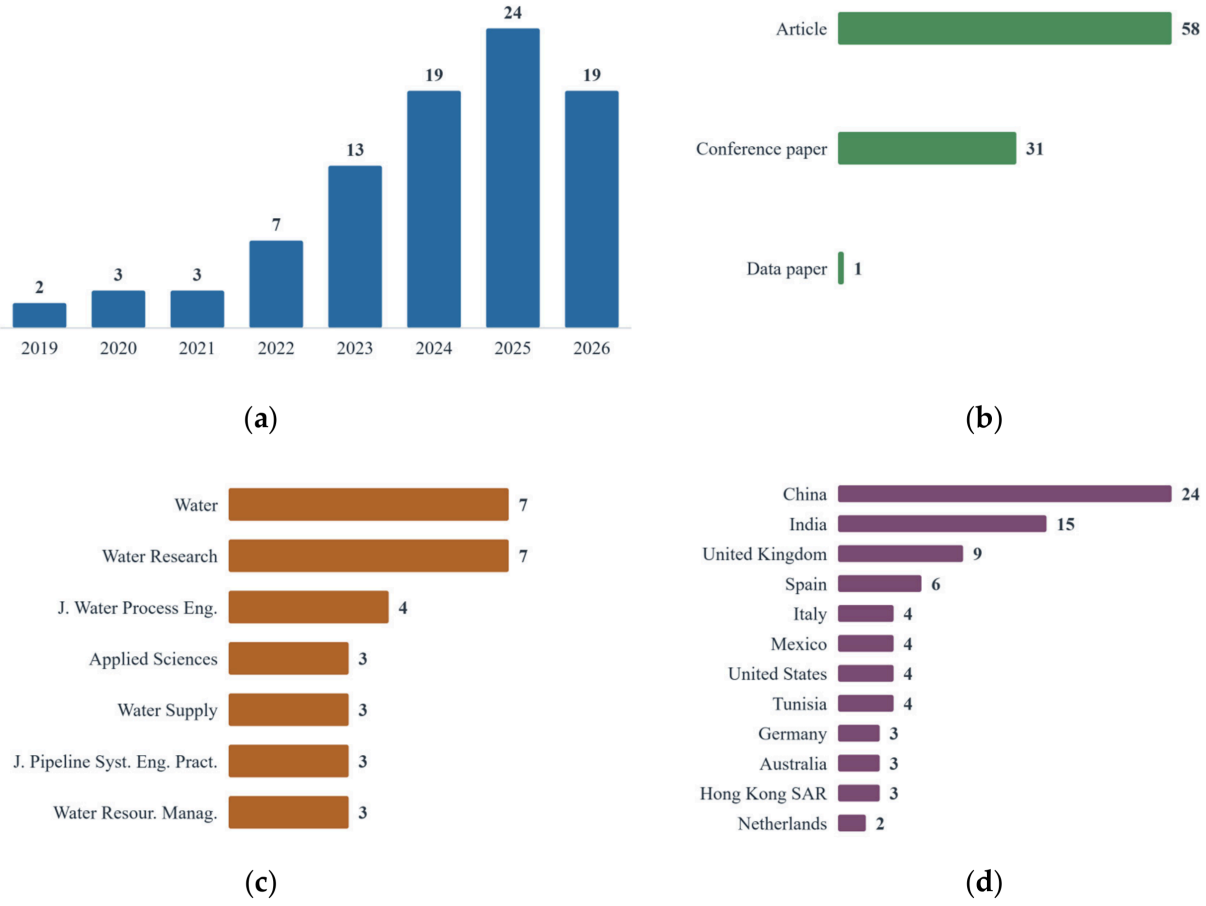


Figure 2. *Cont.*

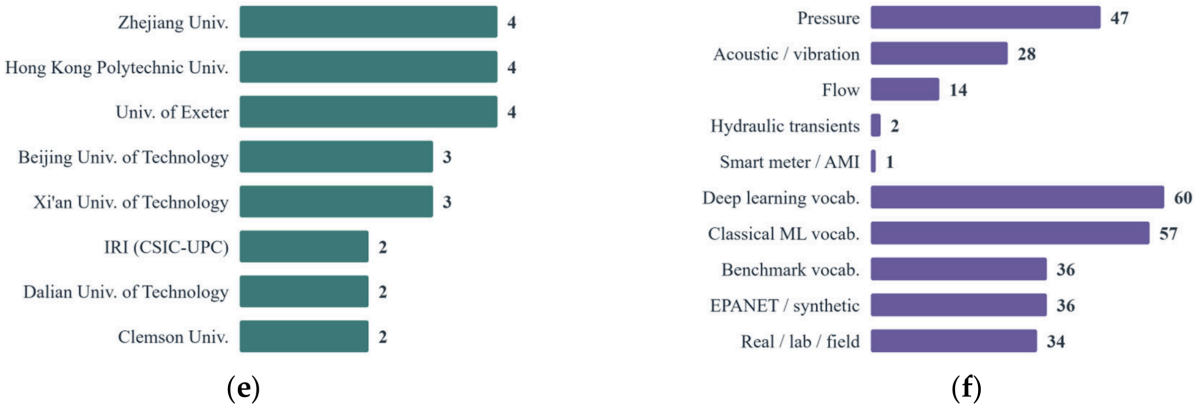


Figure 2. Bibliometric overview of the included studies (n = 90): (**a**) publications per year. (**b**) document types. (**c**) leading publication sources. (**d**) country distribution. (**e**) leading institutions. (**f**) thematic coverage across data modalities and methodological themes. Categories in panel (**f**) are non-exclusive, so a study may appear in more than one row.

3.2. Publication Venues and Impact

The included studies are concentrated in the core water resources and environmental engineering literature: Water (7 studies), Water Research (7), Journal of Water Process Engineering (4), Applied Sciences (3), Water Supply (3), Water Resources Management (3), and Journal of Pipeline Systems Engineering and Practice (3). These are complemented by AI-oriented and instrumentation-oriented venues such as IEEE Access, Engineering Applications of Artificial Intelligence, and IEEE Transactions on Instrumentation and Measurement. Citation impact is highly skewed (mean 13.8, median 4, maximum 110), and 28 studies, predominantly 2025 and 2026 publications, have not yet been cited. The most cited works are, in order, the BattLeDIM summary [19], vibration-signal classification with DL [22], the 1D CNN autoencoder for multivariate SCADA data [8], pressure-based data-driven leak localization [23], AI-based acoustic detection [24], and early deep learning localization on pressure maps [6]. Four of the six most cited papers are benchmark-oriented or comparison-oriented, which indicates a demand within the community for common reference points.

3.3. Geographic and Institutional Distribution

The affiliation analysis (Figure 2d,e) shows a research effort led by Asia. Authors from China contributed to 24 of the 90 included studies (26.7%) and authors from India to 15 (16.7%), followed by the United Kingdom (9), Spain (6), and a second group comprising Italy, Mexico, the United States, and Tunisia (4 each) and Germany, Australia, and Hong Kong SAR (3 each). At the institutional level, The Hong Kong Polytechnic University contributed the most, with 4 studies, followed by Xi'an University of Technology, Zhejiang University, the University of Exeter, and the National Engineering School of Sfax with 3 each, while the Institut de Robòtica i Informàtica Industrial (CSIC-UPC) and Beijing University of Technology contributed 2 each. Two implications for the standardization agenda follow. First, the geography of dataset-aware ML/DL leak detection differs from that of leak detection research in general, in which the United States and the United Kingdom have historically led [2]. The center of activity has moved towards Chinese and Indian groups, many of which work with acoustic or bespoke synthetic data. Second, the leading institutions are distributed across three continents, and no single dominant center is present. This strengthens the case for community benchmarks, since no laboratory can establish a common standard through publication volume alone, and shared infrastructure must therefore be organized explicitly, as was done in the BattLeDIM initiative [19].

3.4. Keyword Co-Occurrence Structure

The author-keyword co-occurrence network (Figure 3), which contains 31 normalized keywords with at least two occurrences and 116 links, shows that the included studies are organized in five modularity clusters. In the largest cluster, the core problem vocabulary (leak detection, water distribution networks) is coupled to the classical ML method vocabulary and to monitoring terms, including machine learning, neural networks, support vector machine, graph neural networks, artificial intelligence, EPANET, signal processing, fault detection, vibration signal, acoustic leak detection, real-time monitoring, and pipeline monitoring. This reflects the development from hydraulic-model-driven detection towards hybrid data-driven approaches [6,23,25]. A second cluster contains the deep learning and localization vocabulary (deep learning, leak localization, LSTM, CNN, water pipeline, feature extraction, fault diagnosis, Fourier transform) together with LeakDB, which confirms that the benchmark, where it is used, is embedded in the deep learning strand of the literature. A third cluster isolates the unsupervised and generative methods (anomaly detection, autoencoders, generative adversarial networks), and two small residual clusters pair sensors with classification and sensor placement with data-driven design. Two structural observations follow for the benchmarking agenda. First, benchmark vocabulary is peripheral. LeakDB is the only named dataset that occurs often enough to enter the network, and no evaluation-protocol terminology (metrics, validation, reproducibility) reaches the occurrence threshold. Second, the acoustic strand is connected to the network mainly through method terms rather than through shared data terms.

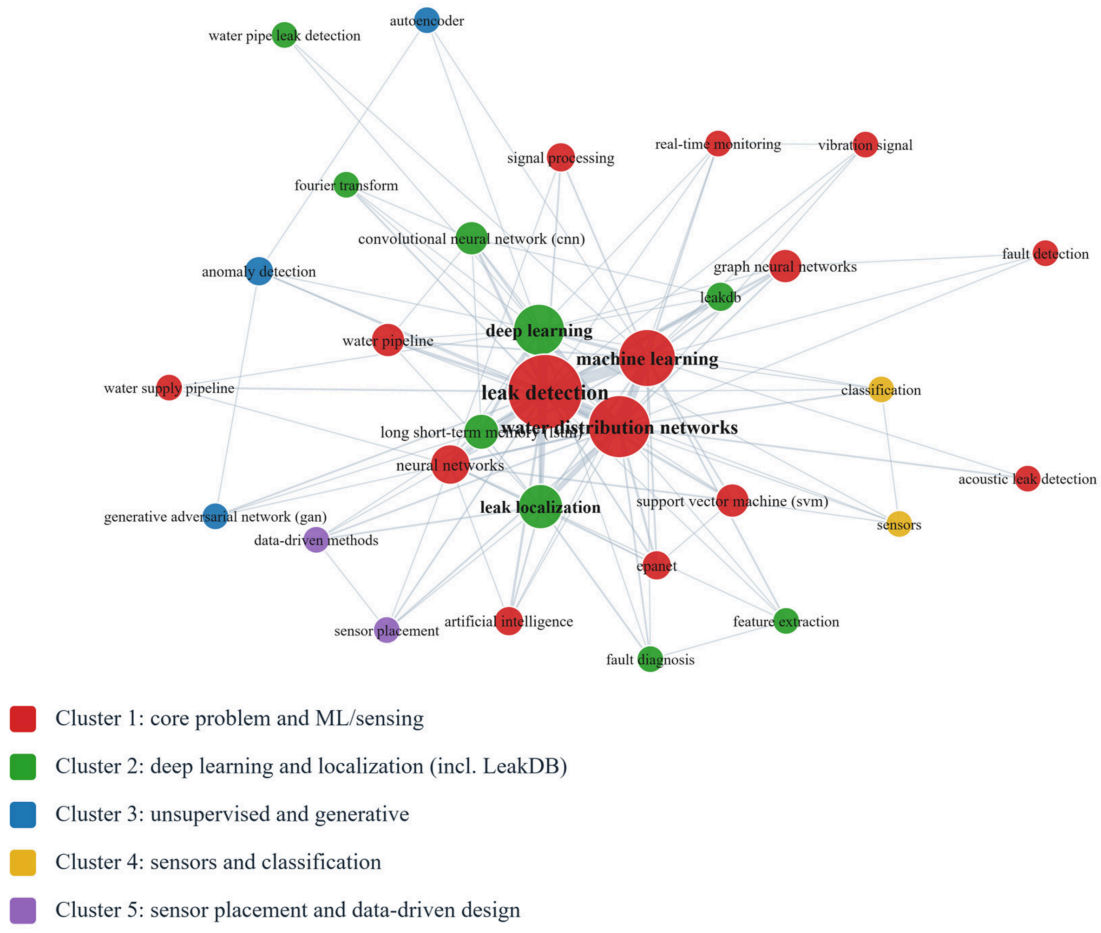


Figure 3. Author-keyword co-occurrence network of the included studies (n = 90). Node size reflects keyword frequency, link width reflects co-occurrence strength, and color reflects modularity cluster membership.

3.5. Methods and Modalities

Deep learning and classical ML appear in comparable proportions, since DL vocabulary is mentioned in 60 studies and classical ML vocabulary in 57, with substantial overlap that reflects hybrid studies. Among sensing modalities, and counting each modality mentioned in the title, abstract, or keywords of a study rather than assigning a single modality to each study, pressure is clearly dominant (47 studies, 52.2%), followed by acoustic and vibration signals (28, 31.1%), flow measurements (14, 15.6%), hydraulic transients (2), and smart-meter or AMI data (1). These counts are non-exclusive, since many studies combine modalities; the assignment of one primary modality per study, obtained from the full text, is reported in Section 4.5. This distribution has direct implications for benchmarking. The principal public benchmarks of the community, LeakDB and L-Town, are pressure and flow simulation datasets, whereas acoustic leak detection, which is the second largest modality, has no comparably adopted public benchmark. Acoustic studies are therefore restricted to private laboratory or field recordings [16,22,24,26,27].

4. Datasets Used in the Included Studies

4.1. Public Benchmarks: LeakDB, BattLeDIM, and L-Town

Only 36 of the reviewed studies (40%) explicitly addressed benchmarking, standardization, public datasets, or reproducibility, and an even smaller number reused established community benchmark datasets. The BattLeDIM/L-Town resource was reused in 12 studies, LeakDB in 7, and the Aghashahi acoustic dataset in 4, so that 23 of the 90 studies reused at least one shared dataset. Among these resources, the BattLeDIM initiative [19] represents the most influential benchmarking effort reported in the literature. Organized in 2020, the competition involved 18 academic and industrial teams and introduced the L-Town virtual water distribution network, which includes two years of SCADA measurements with leak events whose exact timing and location are hidden from participants. In addition, an economic performance metric reflecting operational costs was introduced, enabling the first blind and objective comparison of leak detection and localization methods under a common evaluation framework [19].

Despite its importance, several limitations of BattLeDIM were identified by its developers. Sensor-communication delays and measurement uncertainties commonly encountered in operational systems are omitted from the virtual environment. The full variability of real leakage behavior is not represented by the leak model, indirect water-loss costs are excluded from the scoring, and large, abrupt leaks are favored over smaller or gradually developing events [19]. Stronger and faster pressure or flow deviations are generally produced by large abrupt leaks, thereby increasing the signal-to-noise ratio and reducing ambiguity with normal demand variation. In contrast, weaker changes are produced by slowly developing leaks and can more readily be confounded with operational drift and noise. Consequently, external validity for incipient leaks may be overstated when benchmark performance is weighted toward abrupt events. Although the BattLeDIM economic score has rarely been adopted in subsequent studies, an operational cost of EUR 422,660 $\pm$ 11,553 was reported by Wang [28] for concurrent abrupt- and incipient-leak detection. The broader contrast with the 62.5% of self-evaluated studies reporting headline accuracy above 95% (Section 5.1) should be interpreted descriptively rather than causally. More favorable author-defined scenarios are suggested by this contrast, but the performance difference cannot be attributed to the benchmark leak model alone because the tasks, metrics, and validation designs differ.

The importance of standardized evaluation was further demonstrated by the first unified comparison of multiple ML and DL architectures, published in 2026 [29]. Ensemble learning methods, including Random Forest, XGBoost, and LightGBM, together with Long Short-Term Memory (LSTM) networks, Transformers, and Graph Neural Networks (GNNs), were

evaluated using a common experimental protocol. Detection accuracies exceeding 95% were achieved by all model families, suggesting that binary leak detection under simulated conditions has reached a high level of maturity. However, localization performance remained considerably lower, with the best-performing ensemble models achieving Top-1 localization accuracies between 75% and 77%. In the same study, static GNN architectures were reported to perform poorly in localization tasks, whereas ensemble methods exhibited greater robustness across different leak magnitudes, sensor configurations, and noise levels [29].

These findings are not consistently supported throughout the literature. In a separate investigation using EPANET-generated datasets and a laboratory-scale water distribution network instrumented with 20 pressure sensors, GNNs achieved the highest localization performance, reducing the mean localization error by approximately 50% compared with XGBoost [30]. Because the two studies employed different network topologies, leak scenarios, sensor densities, and localization metrics, their conclusions cannot be directly compared. This discrepancy highlights the lack of standardized evaluation protocols and demonstrates that even fundamental questions regarding the relative performance of different ML architectures remain unresolved.

LeakDB has also been widely adopted as a benchmark dataset for model development and evaluation. Punukollu et al. [31] used LeakDB scenarios generated on the Net1 and Hanoi benchmark networks to compare convolutional neural networks (CNNs) and LSTM models under abrupt, incipient, and mixed leak conditions. Similarly, graph-based methods have been evaluated using the same benchmark topologies [12,14]. Doss et al. [10] proposed a different evaluation strategy by combining the public L-Town benchmark with pressure measurements collected from an operational water distribution network in Norway. Three reconstruction-based architectures, namely Autoencoders (AE), Variational Autoencoders (VAE), and LSTM-AE models, were trained using only leak-free data within a semi-supervised framework. The integration of benchmark datasets with real utility data remains uncommon, although such an approach provides an effective basis for evaluating the transferability of models from synthetic to operational environments.

Only one study within the reviewed literature focused primarily on dataset development. Babela et al. [32] introduced an openly accessible dataset containing one year of hourly SCADA pressure and flow measurements, energy consumption variables, and environmental data collected from a Slovak water utility. Eighteen variables exhibiting statistical relationships with leak events were selected and converted into standardized anomaly scores ranging from 0 to 100. Although the dataset was proposed as a reusable benchmark, several limitations were acknowledged. The predictive capability of conventional pressure and flow measurements was reported to be limited; the dataset represents only a single network zone; and the normalization procedure requires further investigation when additional data are incorporated [32]. Nevertheless, the publication of this dataset represents an important step toward improving the availability of real-world benchmark data for ML-based leak detection.

A related development further supports the need for standardized benchmark datasets. The DiTEC-WDN dataset, published in 2025, contains approximately 36,000 simulated hydraulic scenarios generated across multiple water distribution networks, corresponding to more than 228 million graph-based system states [33]. Unlike existing leak detection benchmarks, the dataset was developed to support large-scale benchmarking under an open-access license. Its developers also identified limitations in LeakDB, particularly the repeated use of demand patterns across multiple nodes, which results in unrealistically high demand correlations [33]. Although DiTEC-WDN does not include leak annotations and therefore cannot directly replace existing leak detection benchmarks such as BattLeDIM or LeakDB, it demonstrates that large-scale, openly accessible, and well-documented

benchmark datasets can be developed. Such datasets provide a valuable foundation for future benchmark standardization and for improving the realism and reproducibility of ML/DL-based leak detection research.

4.2. Bespoke Synthetic Data: The EPANET Default

The most common approach for data generation in the reviewed studies was the use of custom hydraulic simulations. Synthetic datasets generated using EPANET or similar hydraulic modeling tools were employed in 36 studies (40.0%). In most cases, a benchmark network (e.g., Net1 or Hanoi) or a local case-study network was selected [34,35], a hydraulic model was calibrated, and leak scenarios were generated by introducing emitter coefficients with different locations, magnitudes, and initiation times [6,23,25,36]. For example, Gómez-Coronel et al. [25] simulated small leaks in a calibrated EPANET model and applied pressure center-of-gravity displacement for leak localization, whereas Pérez-Sandoval et al. [36] combined network sectorization with deep learning using synthetic leak scenarios to reduce monitoring requirements and dataset size.

This widespread reliance on custom simulation datasets has important implications for the comparability of published results. First, the simulated leak scenarios are defined independently by each study. Consequently, leak sizes, demand patterns, sensor configurations, and noise assumptions differ considerably, making direct comparison of model performance difficult, even when the same ML or DL architecture is evaluated. Second, the level of hydraulic realism varies substantially across studies. Demand uncertainty, sensor noise, and model calibration errors are rarely incorporated at levels representative of real water distribution systems, and uncertainty in model predictions is seldom reported. One notable exception is the framework proposed by Iwakin et al. [37], in which Monte Carlo dropout was applied during inference to estimate confidence intervals for reconstructed sensor values. These uncertainty estimates were subsequently incorporated into a chi-square anomaly detection framework, resulting in reduced false alarm rates on the Net3 network [37]. However, the reported uncertainty reflects only the predictive uncertainty of the learning model rather than the uncertainty associated with the hydraulic simulation itself, and the evaluation was performed exclusively using synthetic data. Third, the use of a common network topology does not necessarily ensure a common benchmark. Studies employing the same hydraulic network may differ substantially in leak definitions, emitter configurations, demand patterns, and sensor placement strategies. As a result, two investigations based on the Hanoi network may be less comparable than studies conducted on entirely different networks. This lack of standardized simulation scenarios limits reproducibility and reduces the value of reported performance comparisons across the literature.

4.3. Laboratory and Field Data

Real-world, laboratory, experimental, and testbed datasets were used in 34 studies (37.8%). Most of these studies focused on acoustic and vibration-based leak detection. Yu et al. [22] classified vibration signals collected using piezoelectric accelerometers from pipe networks located in several Chinese cities, reporting detection accuracies of 95.15% for SqueezeNet and 88.17% for the best conventional classifier. Similarly, Vanijjirattikhan et al. [24] developed an AI-based acoustic leak detection model using field recordings collected from 688 leakage and 619 non-leakage locations.

Several studies combined operational field data with advanced ML techniques. Acoustic datasets collected from real water distribution systems were used by Liu et al. [16] and Xu et al. [26], who incorporated generative data augmentation and model interpretability analysis, respectively. Dunphy et al. [38] employed acceleration measurements to address leak detection under class-imbalanced conditions. Other sensing technologies

have also been investigated, including Fibre Bragg Grating pressure sensors [39], while open-access vibration datasets collected from newly installed water mains have recently become available [40].

At the network level, Kammoun et al. [41] presented one of the few studies that compared multiple ML algorithms under a common laboratory testing protocol. Such controlled experimental evaluations remain uncommon but are essential for ensuring fair and reproducible model comparison. In addition, Tornyeviadzi and Seidu [8] applied semi-supervised learning to multivariate SCADA measurements obtained from operational water distribution systems, where reliable leak labels were not available. This study highlights a fundamental limitation of real-world datasets, as the exact onset and progression of leaks are rarely known in practice. Consequently, ground-truth labels used for supervised learning are often inferred rather than directly observed, introducing uncertainty into both model training and performance evaluation.

4.4. Synthesis: A Divided Data Ecosystem

The datasets used are summarized in Table 2 according to their source, representative applications, strengths, and limitations. Three main categories of datasets were identified. The first consists of publicly available benchmark datasets, such as LeakDB and BattLeDIM/L-Town, which provide standardized data and enable direct comparison between different ML/DL approaches. The second category comprises synthetic datasets generated using EPANET or other hydraulic simulation platforms. Although these datasets offer flexibility in designing leak scenarios and allow the generation of large volumes of labeled data, they are typically developed using study-specific assumptions regarding network configuration, leak characteristics, demand patterns, and sensor placement, thereby limiting the comparability of reported results. The third category includes laboratory experiments and real-world utility datasets. These datasets provide more realistic operating conditions and improve the practical relevance of model evaluation, but they are rarely made publicly available because of confidentiality, operational, or data ownership constraints.

Table 2. Categories of datasets used in ML/DL-based water leak detection research.

Data Category	Included Studies	Strengths	Limitations
Public benchmarks (BattLeDIM/L-Town, LeakDB, Aghashahi)	[10,12,14,19,29,31,42]	Comparability; blind evaluation (BattLeDIM); reproducibility	Mainly simulation-based; limited network diversity; restricted pressure and flow measurements; benchmark aging over time
Bespoke synthetic (EPANET and similar)	[6,23,25,28,36,37,43–55]	Complete ground truth; flexible scenario generation; controlled testing conditions; scalable data production	Author-defined scenarios; uncertain realism; limited transferability to real systems; difficult cross-study comparison
Laboratory and testbed	[22,38,39,41,56–70]	Physical leak representation; controlled experimental conditions; high sensor-level resolution	Small-scale networks; limited topological complexity; testbed-specific effects
Field and utility data	[8,16,24,26,32,71–76]	Real operational conditions; actual demand variability and measurement noise; practical relevance	Uncertain leak labels; restricted access; class imbalance; rarely publicly available
Open data papers	[32]	Reusable operational datasets	Limited to individual utilities; low diversity of operating conditions; emerging availability

4.5. Structure of Extracted Evidence

The methodological structure of the reviewed studies was characterized by data source, sensing modality, model architecture, and validation strategy (Figure 4). When each study was assigned to one primary data source, 30 studies were placed in the public-benchmark category, 21 used laboratory or testbed data, 19 used bespoke synthetic data, 18 used field or utility data, and 2 had insufficiently reported provenance (Figure 4a). Considerable heterogeneity was identified within the public-benchmark category. One of the three water-leak resources tracked in Figure 4d (BattLeDIM/L-Town, LeakDB, or the Aghashahi acoustic dataset) was used in 23 studies. Three studies were assigned to this category because a published network topology was reused without an accompanying named dataset. Other public resources were used in four studies: benchmarks from adjacent domains (an industrial-control-system testbed and a satellite-telemetry anomaly dataset) were used in two, and data from a public repository were reported in two without a named leak-detection benchmark. Separately, extensive topology reuse was observed: Hanoi was used in 15 studies, Net1 or Net3 in 11, and C-Town or D-Town in 4, whereas BattLeDIM/L-Town data were used in 12 studies, LeakDB in 7, and the Aghashahi dataset in 4 (Figure 4d). Only the hydraulic graph is fixed when a topology is reused; leak magnitude, demand patterns, sensor placement, noise assumptions, and evaluation conditions can still differ. Therefore, topology reuse should not be interpreted as reuse of a standardized dataset or as evidence that results are directly comparable.

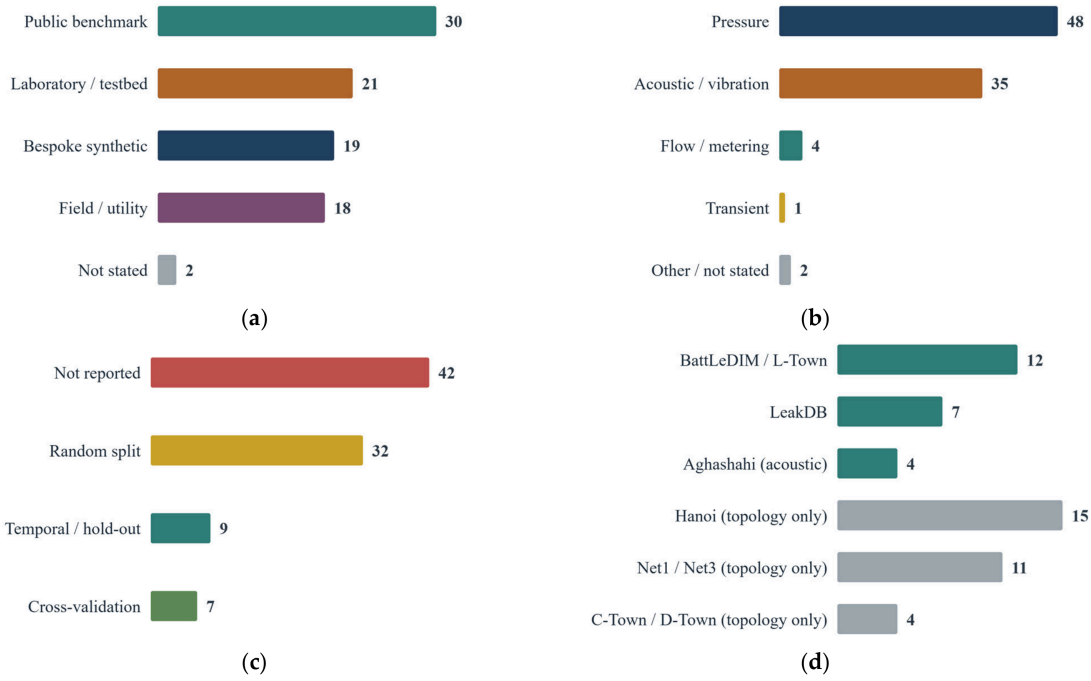


Figure 4. Structure of the evidence base (n = 90 studies): (**a**) primary data source. (**b**) primary sensing modality. (**c**) validation scheme. (**d**) reuse of named datasets.

When a single primary modality is assigned to each study from the full text, rather than counting every modality mentioned as in Section 3.5, pressure is used in 48 studies and acoustic or vibration measurements in 35, followed by flow or metering data (4), hydraulic transients (1), and 2 studies for which the modality is not clearly stated (Figure 4b). The pairing of modality with data source reveals a clear separation between the two dominant research strands. Of the 48 pressure-based studies, 38 rely on public benchmarks (20) or bespoke synthetic datasets (18), which follows directly from the available hydraulic benchmarks being simulation-based. Of the 35 acoustic and vibration studies, 26 rely on

laboratory testbeds (15) or field recordings (11), and only one uses synthetic data. The two principal research directions are consequently evaluated on largely independent data substrates, and no shared resource currently connects hydraulic and acoustic approaches.

Convolutional and feedforward neural networks constitute the most frequently applied model family (33 studies), ahead of recurrent architectures such as LSTM and GRU models (13), classical machine learning and ensemble methods (11), graph neural networks (9), transformer and attention-based models (9), and autoencoder or generative approaches (6). Nine studies do not describe the model family precisely enough to be classified.

Validation practice is the least consistently reported dimension of all (Figure 4c). A random data split is explicitly reported in 32 studies, a temporal or independent hold-out design in 9, and cross-validation in 7. For the remaining 42 studies, the validation procedure is either absent or described too incompletely to establish whether data separation was performed appropriately, for example, where sample counts are given without stating whether leak scenarios generated under the same simulation conditions appear in both the training and testing sets. Because random splitting of pooled synthetic scenarios can introduce information leakage and inflate apparent performance (Section 5.2), the combination of its frequency with this reporting gap is the most consequential finding of the extraction: for nearly half of the reviewed studies, the reader cannot determine whether the reported results reflect performance under genuinely unseen conditions.

The relationship between data source and model family is shown in Figure 5. No data source predominates within any model family, which means that comparisons of architectures drawn across studies are confounded by differences in data provenance. Convolutional and feedforward networks, the largest family, are distributed across field datasets (9 studies), laboratory datasets (9), bespoke synthetic datasets (8), and public benchmarks (7), so differences in reported CNN performance may reflect variation in the data as much as in model design. The more recent architectures are more closely tied to simulation: of the 18 studies applying graph neural networks or transformer-based models, 9 use public benchmarks, 5 use laboratory data, and 4 use bespoke synthetic scenarios, while none use field data. This indicates that advances in model complexity and improvements in data realism are currently progressing along separate pathways.

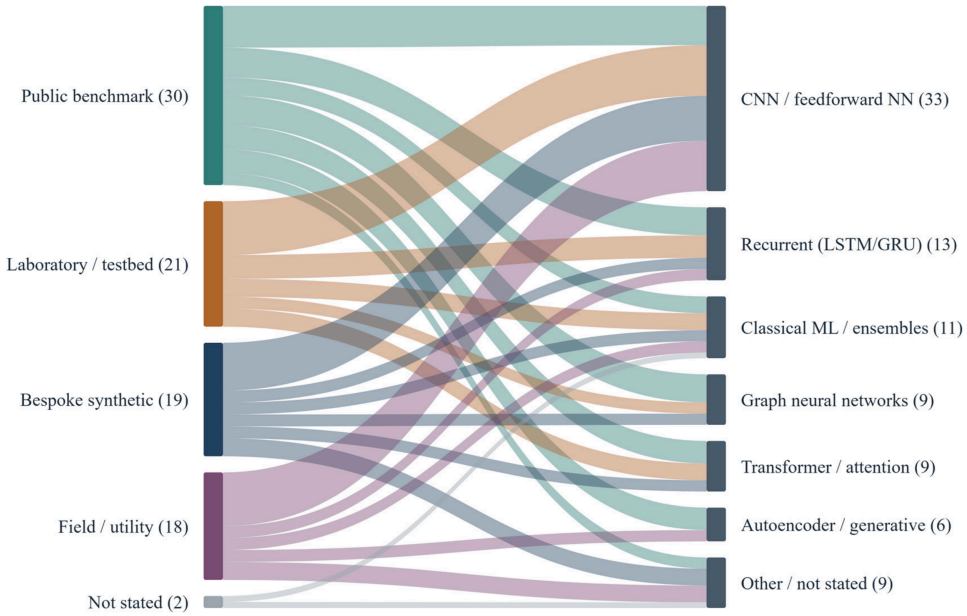
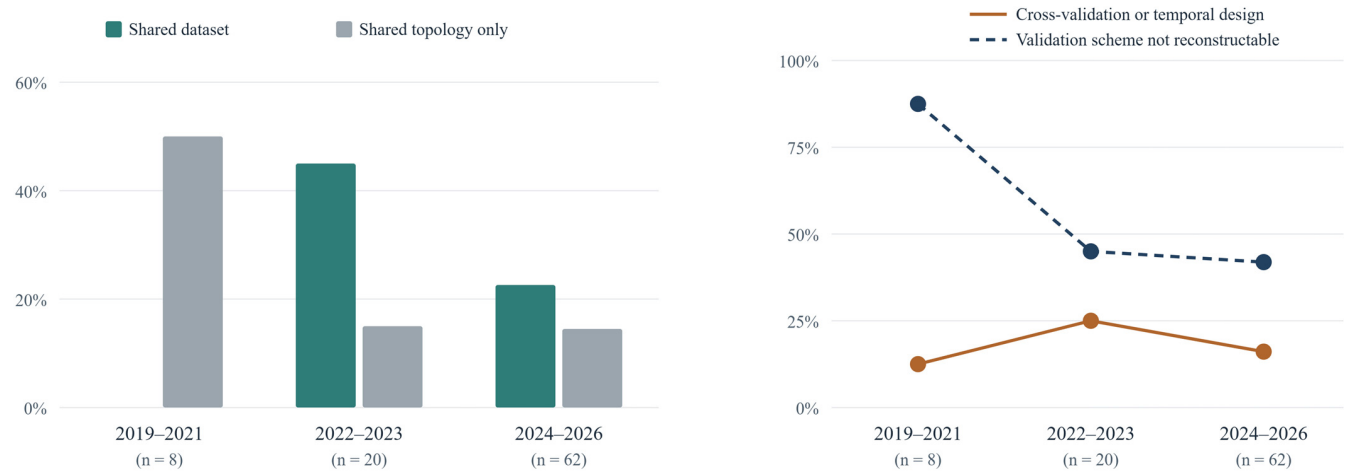


Figure 5. Flow of the included studies from primary data source to model family. Ribbon width is proportional to the number of studies.

4.6. Standardization Practice over Time

Because the publication year was recorded for each study, temporal trends in standardization practices could be assessed rather than inferred (Figure 6). The 90 reviewed studies were grouped into three periods: 2019–2021 (8 studies), 2022–2023 (20 studies), and 2024–2026 (62 studies), reflecting the increasing publication activity observed in Section 3.1. The evolution of standardization practices did not show continuous improvement over time. The reuse of shared benchmark datasets, including BattLeDIM/L-Town, LeakDB, and the Aghashahi acoustic dataset, increased from none of the studies published during 2019–2021 to 9 of 20 studies (45.0%) during 2022–2023. However, this proportion decreased to 14 of 62 studies (22.6%) during 2024–2026. A similar trend was observed for validation strategies based on cross-validation or temporal/hold-out testing, which increased from 1 of 8 studies (12.5%) in the earliest period to 5 of 20 studies (25.0%) in 2022–2023, before decreasing to 10 of 62 studies (16.1%) in the most recent period. The intermediate period corresponds to the BattLeDIM competition and its subsequent influence, during which a shared benchmark dataset and a common evaluation metric were available to the research community. These results indicate that benchmark adoption has not increased proportionally with the rapid expansion of the research field. Although the number of studies using shared datasets increased from 9 to 14 between the second and third periods, the total number of publications increased approximately threefold. This suggests that the impact of existing benchmarking initiatives has not been sufficient to maintain consistent evaluation practices as research activity expanded. In contrast, the proportion of studies using field or utility data increased from 12.5% to 24.2% across the three periods, indicating a gradual shift toward more realistic datasets. However, this progress has not been accompanied by equivalent improvements in shared evaluation frameworks, limiting the comparability of results obtained from increasingly diverse data sources.



(**a**) (**b**)

Figure 6. Standardization practice over time: (**a**) reuse of shared datasets compared with reuse of shared topologies alone; (**b**) validation reporting.

5. Evaluation Protocols and Metrics

5.1. Variability of Metrics

Evaluation practices among the reviewed studies show substantial variability, with no consistent framework adopted across the literature. For leak detection tasks, commonly reported metrics include accuracy, precision, recall, F1-score, false positive rate, false alarm rate, area under the curve (AUC), detection delay, time-to-detection, and reconstruction

error thresholds. For localization tasks, performance is assessed using node-level classification accuracy, top-k ranking accuracy, distance-based localization error, and zone- or sector-level identification. The diversity of evaluation criteria and task definitions limits direct comparison between published results (Table 3).

Several examples demonstrate the impact of dataset characteristics and evaluation conditions on reported performance. Kammoun et al. [41] reported a true positive rate of 90.73%, a false positive rate of 0.13%, and an accuracy of 98.02% for abrupt leak detection using an SVM model on the Net1 network. However, the same study showed a difference of approximately 45 percentage points in true positive rate between abrupt and incipient leaks across all tested algorithms, demonstrating that leak scenario characteristics strongly influence reported performance. Similarly, using the same LeakDB benchmark and the same model architectures, Punukollu et al. [31] reported detection accuracies ranging from 90.56% to 98.23% on Net1 and from 49.00% to 96.55% on Hanoi. This variation highlights the influence of network characteristics and configuration on model performance independently of the selected algorithm.

Other studies have reported high-performance values under different experimental conditions. Damonti and Bernasconi [77] achieved an accuracy of 92.80% (TPR = 91.05%, FPR = 1.66%) using an EPANET-based benchmark, while Dunphy et al. [38] reported laboratory acceleration-based classification accuracies between 98% and 99% with F1-scores ranging from 0.92 to 1.00. Chen et al. [78] reported 99.40% precision and 99.20% recall for acoustic spectrogram classification, whereas Bohorquez et al. [79] obtained an average localization error of 0.59% on a 37.24 m laboratory pipeline. Additional studies focused on hydraulic prediction rather than direct leak classification, such as Momeni and Piratla [80], who reduced pressure prediction MAPE from 3.32% to 2.09%, and Molina et al. [81], who detected a 13.13 L/s flow reduction 120 min after synthetic leak initiation.

These results represent different problem formulations, including binary classification, signal-based detection, localization regression, and hydraulic-state prediction, evaluated using different metrics and data sources. Therefore, they cannot be directly ranked, although they are frequently presented as evidence of improved leak detection performance. This issue is particularly important because leak detection datasets are typically highly imbalanced, with normal operating conditions substantially exceeding leak events [13,38]. Consequently, accuracy alone may provide a misleading assessment of model capability, especially when it is reported without complementary imbalance-sensitive metrics [82].

The BattLeDIM competition addressed this limitation by introducing an economically weighted evaluation metric that integrates detection timing and localization accuracy [19]. Although similar approaches have been adopted in a limited number of subsequent studies, such as the concurrent leak framework proposed by Wang [28], they remain uncommon. Further evidence of the limitations of conventional metrics was provided by the WDSEventDB real-time testbed dataset, where four baseline classifiers evaluated on leakage, cyberattack, and sensor-failure events achieved precision values above 0.92, while recall varied considerably from 0.683 for a multilayer perceptron to 0.271 for a support vector classifier [83]. This demonstrates that high precision values may coexist with poor event detection capability under imbalanced conditions. When evaluation is performed using realistic datasets and metrics sensitive to class imbalance, reported performance is considerably lower than the values commonly obtained from controlled experiments.

Another important limitation identified is the lack of statistical evaluation of model comparisons. Among the reviewed studies, only the acoustic noise-logger comparison [84] investigated whether observed differences between models were statistically significant. Although one-way ANOVA indicated a significant effect of model selection ($p < 0.0001$), pairwise McNemar testing did not confirm that any individual model significantly out-

performed the others (p = 0.0987). Most studies reporting performance differences did not include statistical testing to support their conclusions.

The distribution of reported performance is summarized in Figure 7. Of the 40 studies in which a headline accuracy was stated, values above 95% were reported in 25 (62.5%), values above 99% were reported in 9, and a median of 97.2% was obtained. Median accuracies of 99.0%, 96.7%, 96.5%, and 94.8% were obtained for laboratory and testbed data, field data, public benchmarks, and bespoke synthetic data, respectively. An exploratory Kruskal–Wallis test was applied, and an overall difference among the four distributions of reported accuracy was identified ($H(3) = 8.67$, $p = 0.034$). However, none of the six pairwise comparisons remained significant after Bonferroni correction (corrected $\alpha = 0.0083$); an uncorrected p-value of 0.026 was obtained for the laboratory-versus-field comparison. Because only 7 to 12 studies were included in each group and heterogeneity was present in task, model, validation design, and scenario difficulty, a causal effect of data source cannot be isolated, and no pairwise ranking is supported. Instead, the need for caution when heterogeneous headline accuracies are compared is reinforced. In particular, performance obtained under controlled laboratory conditions may not be generalizable to operational networks with less favorable signal-to-noise ratios and less certain leak labels.

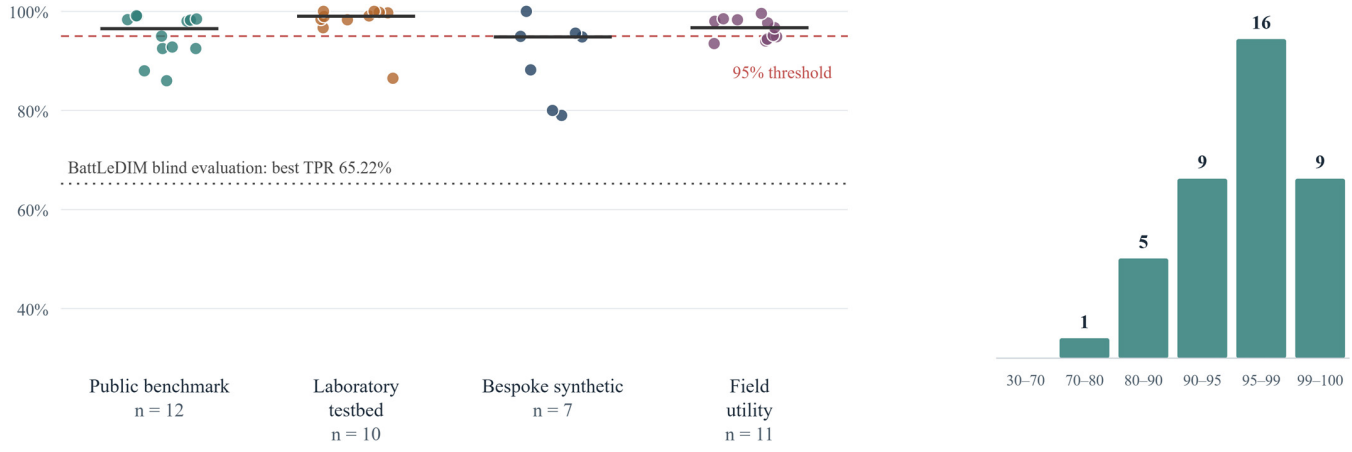


Figure 7. Reported performance across the included studies: (**a**) highest accuracy stated per study against primary data source, with the number of studies given under each group and group medians shown as horizontal bars, for the 40 studies that state a headline accuracy; (**b**) distribution of those values. The dashed line marks the 95% threshold and the dotted line the best BattLeDIM result for previously unseen events (65.22% TPR), obtained under blind evaluation.

This limitation becomes particularly evident when comparing self-reported evaluations with blind benchmarking. In the BattLeDIM competition, which represents the only blind evaluation identified in the reviewed studies, the highest true positive rate achieved for previously unseen leak events was 65.22% [19]. Therefore, reported near-perfect accuracies in the literature should primarily be interpreted as reflecting the characteristics of study-specific evaluation datasets rather than the expected performance under real operational conditions.

Table 3. Evaluation practice extracted from selected studies, illustrating the lack of comparability of metrics across studies.

Study	Data Substrate	Task	Metrics Reported	Headline Result	Validation Scheme
Wang [28]	L-Town-type, concurrent leaks	Detection and localization	Economic score, TP/FP	EUR 422,660 ± 11,553; abrupt and incipient modules complementary	Not reported
Dunphy et al. [38]	Laboratory acceleration signals	Detection and classification	Accuracy, precision, recall, F1	98% to 99% accuracy; F1 0.92 to 1.00; window-slicing augmentation for imbalance	Random split
Kammoun et al. [41]	Net-1 synthetic (pressure/flow)	Detection (abrupt vs. incipient)	TPR, FPR, accuracy	SVM: 90.73% TPR, 0.13 FPR, 98.02% accuracy; 45 percentage points TPR gap from abrupt to incipient	Not reported
Damonti and Bernasconi [77]	EPANET, 500/1000 scenarios	Detection and localization	Accuracy, TPR, FPR, TNR	92.80% accuracy (500 scenarios); GRU/LSTM baselines approx. 48%	Temporal/ hold-out
Chen et al. [78]	Acoustic recordings, Mel spectrograms	Signal classification	Precision, recall	99.40% precision, 99.20% recall	Not reported
Bohorquez et al. [79]	Laboratory transients, 37.24 m pipeline	Localization and sizing	Location error (%), NRMSE	0.59% average location error; NRMSE 2.06%; optimum training-noise intensity	Random split
Momeni and Piratla [80]	Pressure monitoring, hydraulic model	Leakage detection and assessment	MAPE, PCC	MAPE 3.32% to 2.09%; PCC 0.53 to 0.998 across sensor configurations	Not reported
Molina et al. [81]	Synthetic network (Bayesian)	Early detection, simultaneous leaks	Detected flow reduction, time	13.13 L/s reduction detected at min 120	Not reported
Gaurav and Rathi [82]	EPANET simulation	Detection	Accuracy only	RF 88.81%, SVM 88.19%, KNN 87.93%	Not reported
El-Zahab et al. [84]	Field acoustic noise loggers	Detection	Accuracy; ANOVA; McNemar	Ensemble 94.40%; RF 93.68%; significance testing included	Cross-validation
Luo et al. [85]	Hanoi and Net1 synthetic pressure	Detection		Transformer above autoencoder baseline on both networks	Temporal/ hold-out
Meng et al. [86]	Pipeline signals (field)	Detection	Accuracy, recall, F1 (change vs. baselines)	+3.64/+1.66/+1.55 percentage points over MTCNN	Not reported
Zhang et al. [87]	Acoustic spectrograms (non-ML Bi-CFAR)	Detection	Accuracy, recall	97.67%/99.85% ideal; 89.27%/90.07% complex conditions; no training data	Not reported
Chatterjee et al. [88]	Simulated and real DMA data	Early detection	Accuracy, recall, time-to-detect	90% (simulated) vs. 98% (real) acc.; detection within 8 to 12 h of onset	Random split

5.2. Validation Schemes and Leak Scenario Definitions

The validation strategies adopted in the reviewed studies varied considerably, ranging from random train–test splits over pooled datasets to leave-scenario-out approaches and temporal validation schemes designed to better represent real-world deployment. Random splitting of pooled synthetic scenarios was the most commonly reported approach; however, this strategy may introduce information leakage when highly similar scenarios are included in both the training and testing sets, leading to overly optimistic performance estimates.

The definition of leak scenarios also varied substantially across the literature, further limiting the comparability of published results. Different studies considered abrupt and incipient leaks [28,31], background leakage and pipe bursts, or single and concurrent leak events, while the corresponding definitions and leak size thresholds were not consistently reported. Consequently, models were often evaluated under different levels of task complexity. Wang [28] explicitly addressed the more challenging scenario of concurrent abrupt and incipient leaks, highlighting that most previous studies focused on isolated single-leak events. This difference in evaluation settings should be considered when comparing the reported performance of ML/DL models.

5.3. Baselines, Ablation Studies, and Comparability

Comparative evaluations conducted under a common experimental protocol remain limited, although several representative examples were identified. Kammoun et al. [41] compared multiple machine learning algorithms using a common laboratory framework, while Doss et al. [10] evaluated different encoder–decoder architectures within the same experimental pipeline. Barrientos-Torres et al. [89] systematically compared wavelet scattering features with manually engineered features, and the unified benchmarking study reported by [29] provided the first comprehensive comparison of conventional ML and DL architectures for leak detection and localization under a standardized evaluation framework. The fact that such a benchmark was only introduced in 2026 highlights the limited adoption of standardized comparative evaluation practices within the field.

In contrast, the majority of the reviewed studies either did not include baseline models, compared methods implemented under insufficiently described settings, or evaluated algorithms using different datasets, making direct comparison difficult. Full-text analysis also revealed several reporting deficiencies that reduce the reproducibility of published work. For example, in the graph convolutional network (GCN) localization study reported in [14], key implementation details, including convolution kernel size, stride, and padding, were not provided, preventing quantitative comparison with other methods. In addition, the sensing configurations adopted in the study were acknowledged by the authors to exceed those typically achievable in operational water distribution systems, thereby limiting the practical applicability of the reported results.

6. The Gap Between Synthetic and Real Data and Data-Efficiency Strategies

6.1. Importance of Synthetic-to-Real Transfer for Benchmarking

The reviewed studies were almost equally distributed between simulation-based datasets (36 studies) and real-world or experimental datasets (34 studies). Nevertheless, direct transfer between these domains was evaluated only rarely. A focused re-audit of the 90 included studies identified four within-study contrasts relevant to domain or condition shift (Table 4). These results were mapped rather than pooled because the studies used different tasks, metrics, datasets, and validation designs.

Table 4. Evidence map of within-study domain or condition-shift contrasts identified in the reviewed corpus.

Study	Contrast	Reported Result	Interpretive Limitation
Doss et al. [10]	Public benchmark and operational utility data	Both domains were evaluated; no matched attenuation estimate was reported.	Cross-domain framework, but no paired transfer design.
Kammoun et al. [41]	Abrupt versus incipient leaks on synthetic Net-1 data	The TPR was 45 percentage points lower for incipient leaks.	Leak-type shift only; no field transfer, and the validation scheme was not reconstructable.
Zhang et al. [87]	Ideal versus complex acoustic conditions	Accuracy decreased from 97.67% to 89.27%; recall decreased from 99.85% to 90.07%.	Non-ML Bi-CFAR method; the conditions were not a synthetic-to-real pair.
Chatterjee et al. [88]	Simulated versus real DMA data	Accuracy was 90% on simulated data and 98% on real data.	Different, non-paired conditions; the difference cannot be interpreted as transfer attenuation.

Table 4 should therefore be interpreted as an evidence map rather than a meta-analysis. Doss et al. [10] evaluated public-benchmark and operational utility data within one framework but did not report a matched attenuation estimate. Kammoun et al. [41] quantified a leak-type contrast, whereas Zhang et al. [87] quantified an ideal-to-complex acoustic-condition contrast; neither constitutes synthetic-to-real transfer. Chatterjee et al. [88] reported simulated and real-data results, but the conditions were not paired and the real-data accuracy was higher, so the difference cannot be interpreted as transfer loss. Damonti and Bernasconi [77] and Luo et al. [85] also identified real-network validation as necessary without reporting a controlled transfer effect. Thus, no eligible study provided a controlled estimate of performance attenuation across matched synthetic and real conditions, networks, noise levels, or leak types. This absence substantiates the cross-domain validation gap and indicates that future studies should predefine paired conditions, a common metric, and a held-out transfer protocol.

The evaluation of synthetic-to-real transfer is particularly important for practical deployment, as ML/DL models developed using simulated data are ultimately intended for application in operational water distribution systems, where network conditions cannot be fully reproduced through hydraulic simulation. This challenge is further emphasized by recent research on model interpretability. Xu et al. [26] observed that most ML-based leak detection studies prioritize model development while giving limited attention to explainability, thereby reducing transparency and confidence in operational applications. To address this limitation, several explainable AI frameworks [90–92] and interpretable fuzzy inference approaches [93] have recently been proposed to improve the practical adoption of data-driven leak detection systems.

6.2. Data Scarcity, Class Imbalance, and Label Uncertainty

The availability of real leak data remains limited, and existing datasets are typically characterized by class imbalance and uncertain leak labels [8,13,27]. Consequently, numerous studies have investigated methods for improving model performance under limited-data conditions. Dunphy et al. [38] addressed severe class imbalance by applying window-slicing data augmentation and identified generative adversarial networks (GANs) and autoencoder-based models as promising approaches for overcoming data scarcity.

A broad range of data-efficient learning strategies has subsequently been investigated. These include conditional deep convolutional GANs for generating pressure-image training samples [7], integrated generative models for augmenting limited acoustic datasets [16], few-shot acoustic leak detection using hierarchical consistency learning [27], unsupervised acoustic denoising methods [94], domain-adversarial transfer learning across utilities [95], contrastive representation learning for pipeline monitoring [86], semi-supervised autoencoder frameworks that reduce dependence on labeled leak events [8,9,11], unsupervised dynamic graph learning for small-sample applications [13], and metaheuristic feature selection approaches capable of maintaining high detection performance under noisy sensing conditions [96].

Although these techniques represent important advances in data-efficient learning, they also introduce additional challenges for benchmarking. Reported improvements may depend not only on the learning algorithm itself but also on the amount of labeled data, augmentation strategy, and sampling procedure employed during training. Therefore, future benchmark protocols should incorporate standardized label budgets, common augmentation procedures, and consistent evaluation settings to enable fair comparison of data-efficient ML/DL approaches.

6.3. Emerging Infrastructure: Digital Twins, Edge Computing, and Foundation Models

Several emerging technologies are expected to influence the next generation of benchmarking frameworks for water leak detection. Digital twin platforms integrate calibrated hydraulic models with real-time SCADA data to support continuous monitoring, leak detection, and operational decision-making [97]. By combining simulated and operational information within a unified environment, digital twins provide an attractive platform for standardized replay-based benchmarking and synthetic-to-real evaluation. These technologies are discussed here as forward-looking perspectives rather than established practice within the reviewed corpus: none of the 90 included studies reports an operational digital twin deployment, and only isolated examples address edge or cloud computing constraints [76] or foundation-model pre-training [17] directly, so their maturity for water leak detection specifically remains low relative to their use in adjacent domains.

Edge and cloud computing architectures [76] introduce additional performance requirements, including communication latency, computational efficiency, and bandwidth constraints, which are not considered in most existing benchmark protocols but are critical for real-time deployment.

Recent developments in foundation models and large language model (LLM)-based frameworks, such as LeakGPT [17], introduce a pre-training and fine-tuning paradigm that requires benchmark datasets to remain independent of pre-training data in order to ensure unbiased evaluation. Similarly, physics-informed neural networks (PINNs), which incorporate governing hydraulic equations into the learning process rather than relying exclusively on observational data [98], require evaluation protocols capable of assessing generalization beyond the training distribution rather than performance under similar operating conditions. Finally, sensor placement optimization [99] should also be considered within future benchmarking frameworks, as model performance depends not only on the learning algorithm but also on the sensing configuration adopted for data acquisition.

7. Towards Standardization: A Minimum Reporting Framework

Based on the findings presented in Sections 4–6, a minimum reporting framework for ML/DL-based water leak detection studies is proposed (Table 5). The framework focuses on the methodological aspects that exhibited the greatest variability across the reviewed literature while remaining sufficiently concise to be adopted as a practical checklist by both authors and reviewers. The recommended reporting items cover dataset provenance, leak scenario characterization, sensing configuration, data partitioning, evaluation metrics, baseline models, uncertainty and robustness analysis, and reproducibility.

Beyond study-level reporting, the findings of this review indicate four priorities for improving benchmarking practices at the community level.

1. Expansion of benchmark datasets: Existing public benchmark datasets, particularly BattLeDIM/L-Town [19] and LeakDB, remain the primary resources for comparative evaluation but are limited to pressure- and flow-based hydraulic simulations. Future benchmark development should prioritize larger and more diverse network topologies, realistic demand variability, sensor uncertainty, concurrent and incipient leak scenarios [28], and additional sensing modalities, particularly acoustic measurements. The publication of recent open datasets, including DiTEC-WDN [33] and WDSEventDB [83], demonstrates that large-scale and openly accessible benchmarking resources are achievable, although both datasets currently address only part of the benchmarking requirements identified in this review.
2. Standardization of synthetic-to-real evaluation: Model performance should be reported according to clearly defined levels of generalization rather than using a single overall accuracy value. Evaluation should distinguish between in-domain validation,

cross-scenario validation, cross-network validation, and synthetic-to-real transfer. Such a framework would provide a more realistic assessment of model robustness and facilitate comparison between studies. Digital twin environments [97] and combined benchmark–field evaluation frameworks [10] provide promising platforms for implementing these evaluation strategies.

3. Adoption of operationally meaningful evaluation metrics: Conventional classification metrics should be complemented by evaluation measures that reflect operational decision-making. The economically weighted scoring framework introduced in the BattLeDIM competition [19], which jointly considers detection timeliness and localization accuracy, provides an appropriate foundation for future benchmark evaluation and should be reported alongside conventional ML performance metrics.
4. Standardized reporting of uncertainty and interpretability: Reliable deployment of ML/DL models requires not only accurate predictions but also transparent decision-making and quantified prediction uncertainty. Therefore, future studies should report uncertainty estimates, robustness analyses, and model interpretability using standardized procedures to improve confidence in operational applications [26,37,90–93].

Table 5. Proposed minimum reporting checklist for ML/DL water leak detection studies.

Dimension	Required Reporting Items
Dataset provenance	Dataset category (public benchmark, bespoke synthetic, laboratory, or field); network topology and size; simulation software and version (for synthetic datasets); utility anonymization procedure (for field datasets).
Leak scenario characterization	Leak type (abrupt, incipient, background, single, or concurrent); leak magnitude range (absolute value and percentage of system demand); leak onset generation method; total number of leak scenarios.
Sensing configuration	Sensing modality; number and location of sensors; sampling frequency; sensor specifications (field/laboratory datasets) or noise model (synthetic datasets).
Data partitioning	Training, validation, and testing strategy (random, leave-scenario-out, or temporal); measures to prevent information leakage; class distribution within each partition.
Evaluation metrics	Definition of all reported metrics; detection delay; localization metric (node-based, top-k, or distance-based); false alarm rate and evaluation period.
Baseline comparison	Comparison with one or more public benchmark datasets (e.g., LeakDB or BattLeDIM/L-Town), where applicable; baseline implementation details and hyperparameter settings. For modalities without an established public benchmark, such as acoustic leak detection at the time of writing, comparison against a documented internal baseline should be reported instead.
Uncertainty and robustness	Performance variability across repeated runs; sensitivity analysis for sensor noise, demand uncertainty, and sensor failures; uncertainty quantification, where applicable.
Reproducibility	Data availability statement; repository link or documented access restrictions; source code availability; software environment and implementation specifications.

The proposed reporting framework and accompanying recommendations provide a practical foundation for improving transparency, reproducibility, and comparability, but their feasibility has not yet been empirically validated. Accordingly, Table 5 is presented as a proposal for prospective testing rather than as an established standard. Adoption may be constrained by journal page limits, data-sharing policies, the absence of supplementary-material mechanisms in some venues, and disincentives to report validation weaknesses or uncertain estimates. A tiered implementation is therefore proposed: dataset category, validation scheme, test-set independence, class balance, and the primary metric should be

reported in the main text or a structured abstract, whereas scenario details, robustness analyses, code and data access, and repeated-run uncertainty may be supplied in a standardized supplement. Pilot use by journals, benchmark organizers, and independent reviewers is needed to assess completion time, inter-rater consistency, and the proportion of items that can be reported across data modalities. Until such testing is completed, the framework should be regarded as a transparent minimum proposal requiring empirical evaluation and refinement.

8. Discussion and Future Directions

8.1. Principal Findings and Limitations

Five major findings emerge from this review. First, the reuse of standardized benchmarks remains limited and represents an exception rather than an established practice. Although benchmarking terminology is reported in 36 studies, the detailed extraction reveals that only 12 studies employed the BattLeDIM or L-Town dataset, 7 relied on LeakDB, and 4 used the Aghashahi acoustic dataset. In several cases, benchmark usage was attributed to the reuse of a common network topology, such as the Hanoi network, while the associated leak scenarios, data generation procedures, and evaluation protocols were independently defined by the authors. The first attempt to establish a unified benchmark framework enabling comparison across multiple architectures was only reported in 2026 [29], highlighting the recent emergence of systematic benchmarking efforts.

Second, a significant imbalance exists within the data ecosystem across sensing modalities. The reviewed evidence indicates that the two dominant research directions have developed using fundamentally different data foundations. Pressure-based approaches have benefited from the availability of shared benchmark datasets, whereas acoustic-based approaches, despite representing a comparable number of studies and being more closely aligned with field implementation [16,22,24,26], have predominantly relied on privately generated laboratory experiments or field measurements. This separation limits direct comparison between methodologies and constrains the assessment of generalization capability across sensing technologies.

Third, validation methodology represents the least consistently reported aspect among the reviewed studies. Among the 90 analyzed publications, only 7 applied cross-validation procedures and 9 adopted temporal or independent hold-out validation strategies. In contrast, 32 studies relied on random data splitting, while 42 either did not report the validation procedure or provided insufficient details for its reconstruction. This issue is particularly critical because random splitting of pooled simulated scenarios may introduce information leakage, leading to overestimated model performance when similar network conditions or leak characteristics are simultaneously present in training and testing datasets.

Fourth, progress toward standardized evaluation practices has not followed a continuous upward trend. The proportion of studies using shared datasets reached its maximum value during 2022–2023, corresponding to the period influenced by the BattLeDIM competition, where 45.0% of studies adopted common datasets. However, this proportion decreased to 22.6% during 2024–2026. A similar trend was observed for more rigorous validation approaches: the combined use of cross-validation and temporal or hold-out strategies increased from 12.5% to 25.0% and subsequently declined to 16.1% (Figure 6). These observations suggest that benchmark adoption was temporarily strengthened by external initiatives but was not maintained as the research field expanded. Several factors plausibly contribute to this decline, although the reviewed studies do not directly investigate the cause, and these should be read as interpretation rather than established fact. A dilution effect is one candidate: the rapid expansion of the field after 2023 (Section 3.1) brought in a larger number of newcomer research groups and venues, which may be less

likely to have engaged with the BattLeDIM or LeakDB benchmarks specifically. Publication pressure in a fast-moving, competitive area may also favor rapid demonstration of a new architecture over the additional effort of preparing and reporting a rigorous validation or benchmark comparison. If the decline continues, the practical implication is that reviewers and venues, rather than the benchmarks themselves, may need to play a larger role in enforcing standardized evaluation, since voluntary adoption appears not to be self-sustaining. Consequently, improvements in evaluation practice appear to depend strongly on sustained community-driven benchmarking efforts rather than isolated competitions.

Fifth, variation in evaluation procedures should be regarded as a fundamental methodological concern rather than a minor reporting inconsistency. The reliability of comparative conclusions is limited by author-defined scenario spaces, validation procedures susceptible to information leakage, and imbalanced datasets evaluated with unweighted accuracy. As shown in Figure 7, a headline accuracy of at least 95% was reported in 62.5% of the applicable studies, and the highest group median was obtained for laboratory datasets. By contrast, a true positive rate of 65.22% for previously unseen leaks was reported in the only blind evaluation identified in the review. Because accuracy and true positive rate are different metrics and the underlying tasks also differ, these values cannot be interpreted as a direct performance gap. Nevertheless, the contrast demonstrates that model claims cannot be separated from dataset design, validation, and metric selection.

Four limitations of this review are acknowledged. First, a single database was used (Scopus). Although Scopus provides broad coverage of the peer-reviewed water-engineering and computer-science literature relevant to this interdisciplinary topic, complementary searches of other databases could enlarge the evidence base. Second, the corpus-level thematic quantification in Section 3 is based on title, abstract, and keyword mapping, whereas the dataset and evaluation analyses in Sections 4–6 rest on the full-text extraction. The two are consistent where they overlap, but the abstract-level figures are coarser and are presented as indicative only. A headline accuracy figure is stated by 40 of the 90 studies, so the performance analysis in Section 5.1 covers that subset rather than the full set, while the structural counts cover all 90. Third, very recent studies from 2025 and 2026 are included whose citation impact and eventual influence cannot yet be assessed. Fourth, no formal risk-of-bias or methodological quality assessment was performed on the included studies; the review instead reports what each study discloses about its data, validation, and evaluation practices, without independently judging the soundness of the underlying research.

8.2. Future Research Directions

Beyond the standardization recommendations discussed in Section 7, several research priorities are identified from the findings of this review.

Future investigations should include: (i) systematic evaluations of the impact of scenario leakage resulting from random splitting strategies applied to simulated datasets; (ii) benchmark frameworks in which sensor placement and leak detection performance are optimized jointly rather than evaluated under predefined sensing configurations [99]; (iii) transfer learning and domain adaptation benchmarks covering multiple utilities, supported by the increasing availability of open utility datasets [32]; (iv) standardized few-shot learning and limited-label evaluation protocols for data-efficient approaches [16,27,86]; and (v) deployment-oriented evaluation criteria incorporating computational latency, communication requirements, and on-device processing capabilities [76].

The establishment of such evaluation frameworks would improve reproducibility, enable more reliable comparison between methodologies, and facilitate the transition of

artificial intelligence-based leak detection approaches from experimental validation toward practical implementation in operational water distribution systems.

9. Conclusions

Substantial advances have been made in water-leak detection through machine learning and deep learning; however, the methodological and evidentiary foundations have not progressed at the same rate as algorithmic innovation. In this systematic review, 102 Scopus-indexed records were examined, of which 90 studies met the inclusion criteria. Limited standardization of datasets, validation procedures, and reporting practices was identified. Bespoke synthetic datasets were used in approximately 40% of the included studies, and a similar proportion was based on private laboratory or utility measurements. Thirty studies were assigned to the public-benchmark category, but heterogeneous resources and some topology-only reuse were included in that category; it should therefore not be interpreted as 30 studies evaluated on a common, directly comparable dataset.

In many studies described as reusing a benchmark, only a shared network topology, such as Hanoi or Net1, was reused, while leak scenarios and evaluation conditions were independently defined. The hydraulic graph is fixed by a topology, whereas the observations, leak cases, and evaluation conditions are also fixed by a complete shared dataset. This distinction should be preserved and reported in future benchmarking initiatives. Across the corpus, reliable comparison of model performance remains precluded by substantial variation in metrics, validation strategies, and leak-scenario definitions.

The implications extend beyond the comparison of individual algorithms. A median accuracy of 99.0% was reported for laboratory studies and 96.5% for public-benchmark studies, whereas a best true positive rate of 65.22% for unseen leaks was reported in the only identified blind competition. Because different metrics and evaluation conditions were used, these figures are not directly comparable; nevertheless, the need for shared tasks, held-out events, imbalance-sensitive metrics, and transparent validation is underscored. In the absence of these controls, apparent differences among model families may reflect the data and protocol as much as the algorithms.

To address these challenges, the minimum reporting checklist proposed in this review, together with the four recommended community actions related to benchmark development, standardized synthetic-to-real evaluation, economically informed performance assessment, and improved interpretability and uncertainty quantification, provides a framework for advancing the field. Adoption of these practices would support the development of a leak detection research ecosystem in which reported improvements are more reproducible, comparable, and representative of the requirements of operational water distribution systems.

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Abbreviations

The following abbreviations are used in this manuscript:

AE	Autoencoder
AI	Artificial intelligence
AMI	Advanced metering infrastructure
ANOVA	Analysis of variance
AUC	Area under the curve
BattLeDIM	Battle of the Leakage Detection and Isolation Methods
CFAR	Constant false alarm rate
CNN	Convolutional neural network
DiTEC-WDN	Digital Twin for Evolutionary Changes in Water Distribution Networks (dataset)
DL	Deep learning
DMA	District metered area
EPANET	Hydraulic and water quality simulation software for pressurized pipe networks
FP	False positive
FPR	False positive rate
GAN	Generative adversarial network
GCN	Graph convolutional network
GNN	Graph neural network
GRU	Gated recurrent unit
KNN	k-nearest neighbors
LeakDB	Leakage Diagnosis Benchmark (dataset)
LeakG3PD	Extended LeakDB simulated dataset generator (dataset)
LLM	Large language model
LSTM	Long short-term memory
LSTM-AE	Long short-term memory autoencoder
MAPE	Mean absolute percentage error
ML	Machine learning
MTCNN	Multi-task convolutional neural network
NRMSE	Normalized root mean square error
PCC	Pearson correlation coefficient
PINN	Physics-informed neural network
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
RF	Random forest
RQ	Research question
SCADA	Supervisory control and data acquisition
SVM	Support vector machine
TNR	True negative rate
TP	True positive
TPR	True positive rate
VAE	Variational autoencoder
WDN	Water distribution network
WDSEventDB	Water Distribution System Event Diagnosis Benchmark (dataset)

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